UNIT 3 · TOPIC 3.2

Wireless Network Security

Quiz

Instructions: This quiz has 18 multiple-choice questions, 1 point each. Circle the letter of the single best answer for each question.

Name: **Date:** **Class:**

1. Which of the following is an example of a managerial control used to govern network security?

- ○ A. Installing a next-generation firewall at the network perimeter
- ○ B. Enabling WPA3 encryption on all wireless access points
- ○ C. Publishing an acceptable use policy that outlines permitted network activities
- ○ D. Running an Nmap scan to inventory active hosts on the network

2. A school district mandates that all staff complete annual network security awareness training before accessing district systems. Which risk-management function does this training most directly serve?

- ○ A. Detecting anomalous network traffic in real time
- ○ B. Encrypting data transmitted across the wireless network
- ○ C. Reducing the likelihood of human-error-based security incidents
- ○ D. Segmenting the staff network from the student network

3. A network manager schedules a periodic review of all wireless access point configurations to verify compliance with the organization's security policy. This activity is best classified as which type of risk-management control?

- ○ A. Corrective control
- ○ B. Detective managerial control
- ○ C. Deterrent control
- ○ D. Technical preventive control

4. A university Wi-Fi network uses WPA2-Personal with a shared passphrase posted in every classroom. A security consultant determines this exposes students to credential theft via offline dictionary attacks. Which change would most directly reduce this risk?

- ○ A. Switching to WPA2-Enterprise with 802.1X per-user authentication
- ○ B. Increasing the shared passphrase length to at least 20 characters
- ○ C. Hiding the network SSID so it does not appear in device scan lists
- ○ D. Enabling MAC address filtering on all wireless access points

5. An organization implements a BYOD policy requiring all employee-owned devices to run approved endpoint security software before connecting to the corporate wireless network. Which of the following best explains how this managerial control reduces insider threat risk?

- ○ A. It blocks unauthorized devices from associating with the access point via MAC address filtering
- ○ B. It ensures devices meet a minimum security baseline, reducing the attack surface available to malicious or negligent insiders
- ○ C. It encrypts all wireless traffic between employee devices and the access point
- ○ D. It automatically segments employee devices onto a dedicated VLAN

6. A network administrator disables SSID broadcast on a guest wireless network as a wireless security measure. Which statement most accurately describes the actual effect of this configuration on the network's risk profile?

- ○ A. It prevents all unauthorized users from discovering the network because unlisted networks are not visible to scanning tools
- ○ B. It qualifies as a managerial control because hiding the SSID is a documented configuration decision
- ○ C. It eliminates the need for WPA2 or WPA3 encryption because undiscoverable networks cannot be targeted
- ○ D. It provides minimal security improvement because the SSID remains discoverable through passive analysis of probe request frames

7. A National Guard technician must configure emergency Wi-Fi at a disaster shelter. Available hardware supports both WPA2-Personal and WPA3-Enterprise. Shelter residents will use personal devices; no on-site RADIUS server infrastructure is available and sourcing one before the shelter opens is not feasible. Evaluating risk-management trade-offs, which configuration best balances security and operational feasibility?

- ○ A. WPA3-Enterprise, because per-user 802.1X credentials eliminate the shared-key vulnerability entirely regardless of infrastructure constraints
- ○ B. WPA2-Personal with the passphrase publicly displayed at the shelter entrance so all residents can connect without assistance
- ○ C. WPA2-Personal with a strong randomly-generated passphrase, combined with guest VLAN segmentation and a posted acceptable use policy for residents
- ○ D. Wi-Fi Enhanced Open (OWE) under WPA3, since it provides opportunistic encryption and removes the passphrase barrier for displaced residents

8. A small company's entire wireless security program consists of a written BYOD policy stating that personal devices must only be used for work tasks. No technical controls, no network access control, no endpoint detection, no wireless intrusion detection, are deployed. Evaluating the effectiveness of this risk-management strategy, which conclusion is most accurate?

- ○ A. The policy is sufficient because managerial controls form the highest tier of the security control hierarchy and set the intent that all other controls serve
- ○ B. The policy is sufficient if all employees sign an acknowledgment form, because that creates enforceable legal accountability for violations
- ○ C. The policy is insufficient because without complementary technical controls, compliance cannot be enforced, monitored, or verified
- ○ D. The policy is sufficient for small companies because threat actors typically focus resources on larger, more profitable targets

9. A network security manager selects risk-management strategies for a hospital's wireless network. Three options are under consideration: Option 1, WPA3-Enterprise encryption only, no written governance policies. Option 2, Written wireless governance policies only, no encryption upgrade from the current WPA2-Personal. Option 3, WPA3-Enterprise encryption combined with a wireless device governance policy. Using NIST SP 800-12 principles, which ranking of these options is most defensible?

- ○ A. Option 1 > Option 2 > Option 3, because technical controls are inherently more effective than managerial ones and combining them introduces policy overhead without additional protection
- ○ B. Option 3 > Option 1 > Option 2, because layered controls address both technical vulnerabilities and behavioral or governance gaps that neither control type resolves alone
- ○ C. Option 2 > Option 3 > Option 1, because governance frameworks establish the authorization structure that gives technical controls their legitimacy and should be prioritized first

○ D. All three options are equivalent for a hospital network because any single well-implemented control, whether technical or managerial, is sufficient given appropriate funding

10. When configuring WPA2 security on a wireless access point, which of the following settings must a technician explicitly configure to enable encryption?

○ A. The network's SSID broadcast name

○ B. The maximum number of simultaneous device connections

○ C. The encryption passphrase or pre-shared key (PSK)

○ D. The radio channel and bandwidth (e.g., 2.4 GHz vs 5 GHz)

11. After changing the wireless encryption standard on a corporate access point from WPA2 to WPA3, which action should the technician take to properly document the mitigation?

○ A. Record the change in a configuration change log with the date, change description, and technician name

○ B. Send an all-staff email announcing that a wireless security upgrade has been completed

○ C. Run a packet capture to confirm that WPA3-encrypted frames are visible on the network

○ D. Reboot the access point to ensure the new configuration is applied

12. A technician wants to implement MAC address filtering on a wireless network. Which of the following correctly describes how this mitigation is configured?

○ A. The technician upgrades the wireless encryption protocol from WPA2 to WPA3

○ B. The technician enters the hardware MAC addresses of permitted devices into an access control list on the wireless access point

○ C. The technician assigns each permitted device a static IP address through DHCP reservations

○ D. The technician reduces the access point's transmit power to limit the wireless signal's physical coverage area

13. A National Guard technician is configuring a new consumer-grade wireless access point for a disaster shelter. The following steps must be completed: (I) Set a strong WPA2-Personal passphrase. (II) Change the default administrator password on the access point management interface. (III) Disable WPS (Wi-Fi Protected Setup). (IV) Enable the SSID so residents can discover and connect to the network. Which sequence correctly reflects secure implementation practice?

○ A. IV → I → III → II

○ B. III → IV → I → II

○ C. I → II → IV → III

○ D. II → I → III → IV

14. A technician configuring WPA2 on a corporate wireless network must choose between the TKIP and CCMP (AES) cipher suites. Which cipher suite should be selected and why?

○ A. TKIP, because it was the original WPA cipher and provides the broadest compatibility with legacy wireless devices on the network

○ B. CCMP (AES), because TKIP has documented cryptographic weaknesses and CCMP provides stronger, standards-compliant encryption

○ C. TKIP, because it uses a 256-bit key, making it cryptographically stronger than CCMP's shorter key

○ D. CCMP (AES), but only if the network has more than 50 concurrently connected devices, since AES adds processing overhead on smaller networks

15. A technician upgrades a hospital's wireless network from WPA2-Personal to WPA3-Enterprise. Which of the following log entries most completely fulfills the "log mitigations" requirement for this configuration change?

- ○ A. "Wi-Fi upgraded."
- ○ B. "Change made to wireless settings per IT manager instruction."
- ○ C. "WPA3-Enterprise enabled on all access points on 2024-11-15 by J. Kim; prior configuration: WPA2-Personal with PSK; reason: policy compliance requirement and elimination of shared-key authentication risk; verified via successful 802.1X authentication test on two client devices."
- ○ D. "WPA3 is now active across the hospital network."

16. A National Guard technician must deploy and document emergency shelter Wi-Fi under the following constraints: consumer-grade access point only, no RADIUS server, no additional hardware procurement time. Applying NIST SP 800-12 principles for implementing and logging protective controls, which combination of mitigations and documentation most appropriately addresses this situation?

- ○ A. Deploy WPA2-Personal with a strong randomly-generated passphrase, disable WPS, apply guest VLAN segmentation, post a written network acceptable use policy, and log each configuration step with timestamp, action taken, and rationale
- ○ B. Deploy WPA3-Enterprise and document that no RADIUS server was required because WPA3 is inherently self-authenticating in Enterprise mode
- ○ C. Deploy the network with no encryption to ensure all displaced residents can connect without technical barriers, and create a detailed post-deployment usage policy
- ○ D. Deploy WPA2-Personal using factory default settings to minimize setup time, and log the decision to prioritize rapid deployment over security configuration

17. As part of a NIST-aligned change management process, a technician simultaneously implements two controls: (1) a new network governance policy mandating that all wireless access points use WPA3, and (2) the technical upgrade of all APs to WPA3-Personal. When logging these mitigations, what should the log entry capture to be most useful for a future security audit?

- ○ A. Both the policy implementation and the technical change, with cross-references between them, dates, technician and approver names, prior configuration state, verification method, and any devices that could not be upgraded
- ○ B. Only the technical change, because governance policies are maintained in a separate policy register and do not belong in a configuration change log
- ○ C. Only the policy, because the technical change will be independently detected and recorded by network monitoring tools during routine traffic analysis
- ○ D. A single summary statement confirming that WPA3 is now both policy and deployed, since brevity improves log readability and auditability

18. A security auditor reviews the following wireless configuration change log entry: "Changed encryption on APs to WPA3. — J. Kim, 2024-11-15." The auditor flags the entry as insufficient for audit purposes. Which set of missing elements, if added, would bring this entry into compliance with NIST SP 800-12 documentation standards for logged mitigations?

- ○ A. The prior configuration state, the justification or reason for the change, the name of the authorizing approver, and verification evidence that the change was successfully applied
- ○ B. Only the time of day is missing; date, technician name, and action described are sufficient for a minor configuration change
- ○ C. Only the approver's name and signature are missing; all other required fields are present in the current entry

○ D. No elements are missing; a concise log entry is acceptable under NIST guidance for routine low-risk configuration changes